Supervisor planning meeting

CHRONO-Fair

Time-Resolved Counterfactual Fairness Monitoring for Clinical Machine Learning

Extending our prior VFR-Audit three-axis static auditing framework

Target venue: Q1 digital-health / health-informatics journal

Where clinical ML stands today

- Clinical ML supports triage, risk scoring, and length-of-stay prediction across many sites [1, 2].
- Documented post-release subgroup failures across radiology, sepsis, kidney function, and care-allocation models [3, 4, 5, 6].
- Regulatory expectations have shifted toward continuous lifecycle monitoring [7, 8] with reporting standards [9, 10, 11, 12].
- Auditing toolkits in routine use return a thresholded pass-or-fail verdict on a fixed snapshot [13, 14, 15, 16].
- That verdict is the object hospitals act on. The snapshot tells us nothing about its stability over time.

Our prior VFR-Audit study (static auditing framework)

Three audit-reliability axes on real Texas-100X (925 128 records) [17]:

- Axis 1: resampling fluctuation of the fairness verdict.
- Axis 2: audit-size sensitivity of the fairness verdict.
- Axis 3: cross-hospital-site agreement of the fairness verdict.

Standard vs Fair monitored-model setup (cited from prior work):

Attribute	Std DI	Fair DI	Verdict
Race	0.654	0.884	fail $\rightarrow$ pass
Sex	0.762	0.933	fail $\rightarrow$ pass
Ethnicity	0.832	0.962	pass $\rightarrow$ pass
Age Group	0.291	0.801	fail $\rightarrow$ pass

Std Acc 0.8777 $\rightarrow$ Fair Acc 0.8248 (reduction of 5.28 pp). Pre-release adjustment is

The remaining gap

Real Texas-100X anchor finding (925 128 records) [17]

The worst-group true-positive-rate verdict for race reads “fair” at **0.824**. The same verdict flips to “unfair” in **4 of 20** hospital-network resamples (Verdict Flip Rate **0.20**). A batch audit cannot tell when, or whether, that verdict reversal recurs as new patients arrive.

Four gaps follow:

- **Gap 1, time axis.** Batch verdicts give no timestamp [13, 14, 15].
- **Gap 2, peeking penalty.** Repeated inspection inflates false-alarm rates without a sequential guarantee [18, 19, 20].
- **Gap 3, diagnostic interpretation.** A detected gap with no triage does not tell a team whether to fix data or collect more samples [21, 22].
- **Gap 4, regression output.** Length-of-stay outputs lack a flip-rate

Closest related work, named and positioned

Tool / work	Mode	Output	Closest gap
Aequitas [13]	Batch	Per-attribute parity-ratio report	No streaming alarm
Fairlearn [14]	Batch	Group fairness metrics; mitigator	No verdict-flip monitoring
AIF360 [15]	Batch	20 fairness metrics library	Snapshot only
Microsoft RAI Dashboard [16]	Notebook batch	Fairlearn + interpret + CF	Developer review, not gove nance
Davis et al. [2]	Quarterly	Subgroup trend, 11 yr VA	No per-cell sequential alarm
Prediction sensitivity [22]	Per batch	CF-flip rate scalar	No alarm rule
VFR-Audit (our prior) [17]	Static	Verdict Flip Rate scalar	No time axis

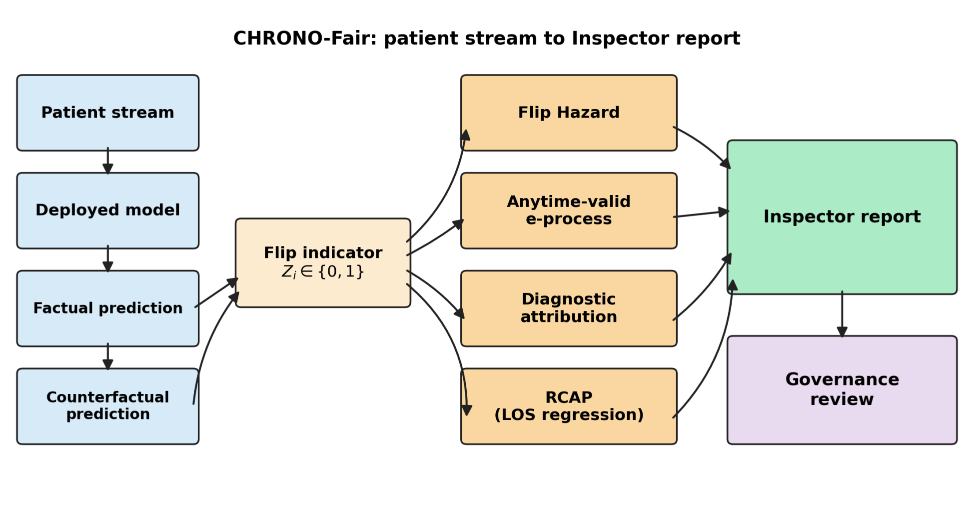
Anytime-valid machinery exists [18, 19, 20] but has not been applied to counterfactual fairness verdicts at intersectional granularity.

Our planned extension, in one sentence

CHRONO-Fair contributes a **post-release, time-resolved monitoring layer** for thresholded fairness verdicts using a **per-cell anytime-valid e-process** on counterfactual flip indicators.

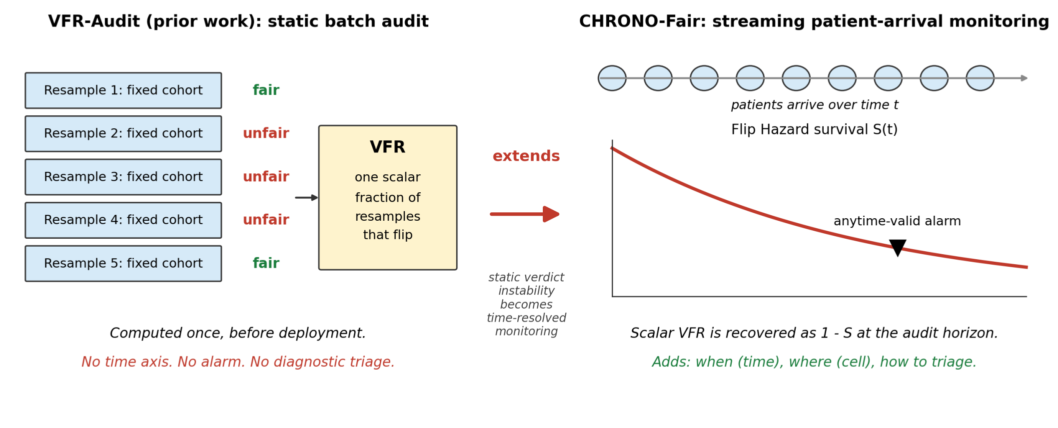
Monitor, not mitigator • Begins after Std $\rightarrow$ Fair • Agnostic to mitigation method • Research prototype, not medical device

Planned architecture



Modules: Flip Hazard [24]; per-cell shrunk-GROW e-process [18, 19]; aleatoric/epistemic

How CHRONO-Fair extends VFR-Audit



Scalar VFR is recovered as $1 - S$ at the audit horizon. CHRONO-Fair adds the time axis, the anytime-valid alarm rule, and the diagnostic triage signal.

Mathematical content: what is taken vs what is ours

Equation / construct	Attribution
Flip indicator Z_i	Our definition; same idea as prediction sensitivity [22]
Flip Hazard $\lambda_a(t)$	Standard hazard form (Kaplan-Meier [24]; Nelson-Aalen 1972); our naming and application
RMFT integral	Standard Restricted Mean Survival Time; adopted
e-process wealth $E_t = \prod(1+\lambda_s(Z_s-\rho_0))$	Waudby-Smith and Ramdas [29]; Ramdas et al. [20]; adopted
Shrunk GROW bet λ_s	Our finite-sample shrinkage of the GROW rule of [29]
Ville's inequality (Theorem 1)	Ville 1939 [18]; standard application
Ensemble flip probability $\bar{p}^{\mathrm{flip}}$	Our definition
Aleatoric / epistemic entropy split	Depeweg et al. [30]; Hullermeier and Waegeman [21]; adopted
RCAP rank positions u_i	Standard nonparametric group-conditional CDF; our naming
RCAP $= W_1(\ldots)$	Wasserstein-1 from optimal transport; adopted

The novelty is not a new survival estimator or a new e-process. The contribution is the coordinated integration of these instruments around the thresholded fairness verdict.

How each planned module closes one identified gap

- **Gap 1, time axis** $\rightarrow$ **Flip Hazard** (per-arrival survival curve instead of a single scalar [24]).
- **Gap 2, peeking penalty** $\rightarrow$ **Per-cell anytime-valid e-process** (Type-I control at every inspection step [18, 19, 20]).
- **Gap 3, diagnostic interpretation** $\rightarrow$ **Aleatoric/epistemic decomposition** (triage label per alarmed cell [21]).
- **Gap 4, regression output** $\rightarrow$ **RCAP** (counterfactual rank shift for LOS regression [23]).

Method discipline: what we will not claim

- **Per-cell anytime-valid Type-I control:** YES, under Ville inequality [18, 19].
- **Step-wise Benjamini-Hochberg across cells:** applied at inspection time only. NOT a time-uniform online FDR guarantee [25, 26].
- **Diagnostic attribution:** aleatoric/epistemic triage signal. NOT a confirmed causal root cause [21].
- **Counterfactual definition:** feature-shift, computable at inference. Weaker than SCM-based counterfactual fairness [27, 28]; may under-flag indirect effects.
- **Clinical deployment:** controlled replay and simulation only. NOT prospective EHR deployment.

Anchor streaming result (controlled replay)

Method	Detected	Mean delay (patients)
soft CHRONO-Fair (per-cell e-process)	**50 / 50**	**837 (median 828)**
ADWIN ($k = 1$)	21 / 50	3 878
soft ADWIN ($k = 25$)	21 / 50	3 927
Davis-style ADWIN on fairness gap	21 / 50	3 904
soft Periodic batch ($n = 500$)	24 / 50	499 (no peeking control)
Static VFR	0 / 50	–

10 000-patient stream, drift onset at patient 5 000, 50 Monte-Carlo runs. Empirical false-alarm rate at or below the nominal α for every tested α [19, 29].

Streaming evidence is controlled replay and simulation. Not prospective clinical validation.

Plan: how the gap will actually be closed

- **Phase 1.** Controlled-validation evidence: real Texas-100X batch + Texas-100X-calibrated streaming/replay + robustness stress tests. *Drafted, in revision for Q1 submission.*
- **Phase 2.** Methodological strengthening: time-uniform online FDR [25, 26]; SCM counterfactual sensitivity [27, 28]. *Planned for next revision.*
- **Phase 3.** Prospective evaluation: site-partnered EHR-stream validation; governance-user evaluation of the Inspector report. *Scoping, partner search.*
- **Phase 4.** Artifact release: anonymised proof-of-concept prototype, configs, seeds, scripts. Subject to anonymisation and journal artifact-release requirements. *Scheduled for camera-ready.*

Asks for this meeting

1. Is the post-release monitoring framing the right contribution to lead with, vs a methods-only framing of the per-cell e-process?
2. Is the Standard / Fair monitored-model setup table (cited from the prior VFR-Audit study) acceptable to use as the predecessor without re-running it?
3. Which Q1 venue is best: Frontiers in Digital Health, npj Digital Medicine, JAMIA, JBI, or NEJM AI?
4. Should I add a second real cohort before submission (MIMIC-IV / eICU), or submit with single-dataset honesty plus a prospective-validation roadmap?
5. Defer artifact release to camera-ready, or supply a minimum-viable artifact preview with the submission for reviewer reproducibility?
6. Timing: next-round submission in 6 to 8 weeks, or invest 2 to 3 months in Phase 2 methodological strengthening first?

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